

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin \frac{x}{a} + c$$

$$\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln |x + \sqrt{x^2 \pm a^2}| + c$$

$$\int \frac{p(x) dx}{\sqrt{ax^2 + bx + c}} = \tilde{p}(x) \sqrt{ax^2 + bx + c} + \int \frac{c}{\sqrt{ax^2 + bx + c}} dx, \text{ st. } (\tilde{p}) = \text{st.}(p) - 1$$

5 a) $\int \frac{1}{\sqrt{x+x^2}} dx = \int \frac{dx}{\sqrt{(x+\frac{1}{2})^2 - \frac{1}{4}}} = \ln(x + \frac{1}{2} + \sqrt{x^2 + x}) + c$

b) $\int \frac{dx}{\sqrt{2x-x^2}} = \int \frac{dx}{\sqrt{1-(x-1)^2}} = \arcsin(x-1) + c$

c) $\int \frac{x+3}{\sqrt{x^2+4x}} dx = A \sqrt{x^2+4x} + \int \frac{C}{\sqrt{x^2+4x}} dx =$

$$\frac{x+3}{\sqrt{x^2+4x}} = A \frac{2x+4}{2\sqrt{x^2+4x}} + C \frac{1}{\sqrt{x^2+4x}}$$

$$2A + C = 3$$

$$A = 1$$

$$C = 1$$

$$= \sqrt{x^2+4x} + \int \frac{dx}{\sqrt{x^2+4x}} = \sqrt{x^2+4x} + \int \frac{dx}{\sqrt{(x+2)^2 - 4}} =$$

$$= \sqrt{x^2+4x} + \ln(x+2 + \sqrt{x^2+4x}) + c$$

Določeni integral

1) Izračunaj z Riemannovo vsoto

$$\int_0^1 x^2 dx = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{n} \cdot \left(\frac{k}{n}\right)^2 = \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=0}^n k^2 = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{2}{6} = \frac{1}{3}$$



$$\Delta x = \frac{1}{n}$$

$$t_i = \frac{k}{n}$$

$$f(t_i) = \left(\frac{k}{n}\right)^2$$

2.

a)
$$\int_0^{2\pi} \cos^2 x dx = \int_0^{2\pi} \frac{1 + \cos 2x}{2} dx = \left. \frac{1}{2}x + \frac{1}{4} \sin t \right|_0^{2\pi} = \pi$$

$$2x = t$$

$$2dx = dt$$

b)
$$\int_0^{\pi} \sqrt{\frac{1 + \cos 2x}{2}} dx = \int_0^{\pi} \cos x dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x dx = \sin x \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 1 - (-1) = 2$$



Trapezna metoda

$$\int_a^b f(x) dx = \frac{b-a}{n} \left(\frac{y_0 + y_n}{2} + y_1 + \dots + y_{n-1} \right) + R_n$$

$$|R_n| \leq \frac{(b-a)^3}{12n^2} \left(\max_{[a,b]} |f''(x)| \right)$$

3. S trapezno metodom izračunaj, $n=4$

$$\int_0^{\pi} \frac{\sin x}{x} dx = \frac{\pi}{4} \left(\frac{1+0}{2} + 1,84 \right) + R_4 = \frac{\pi}{4} (2,34) + R_4 = 1,84 + R_4$$

x_i	y_i
0	1
$\frac{\pi}{4}$	$\frac{\sqrt{2} \cdot \pi/2}{2 \cdot \pi} = \frac{2\sqrt{2}}{\pi} \approx 0,9$
$\frac{\pi}{2}$	$\frac{2}{\pi} = 0,64$
$\frac{3\pi}{4}$	$\frac{\sqrt{2} \cdot \pi/2}{2 \cdot 3\pi} = 0,3$
π	0

4.

a) $\int_0^a \frac{1}{\sqrt{a^2 - x^2}} dx = \quad a > 0$

$$= \lim_{\varepsilon \rightarrow 0} \int_0^{a-\varepsilon} \frac{dx}{\sqrt{a^2 - x^2}} = \lim_{\varepsilon \rightarrow 0} \arcsin \frac{x}{a} \Big|_0^{a-\varepsilon} = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

b) $\int_0^1 \ln x dx = \lim_{\varepsilon \rightarrow 0} \left(x \ln x \Big|_{\varepsilon}^1 - \int_{\varepsilon}^1 dx \right) = -1$

c)
$$\int_0^{\infty} x^n e^{-x} dx = \lim_{n \rightarrow \infty} \left(\underbrace{-x^n e^{-x}}_0 + \int_0^n n x^{n-1} e^{-x} dx \right) = \dots = \lim_{n \rightarrow \infty} \int_0^n n! e^{-x} = n!$$

Kriterij za konv. izlim. integralov

• $\int_a^b \frac{g(x)}{(b-x)^s} dx$

$s < 1 \rightarrow$ konvergira

$s \geq 1 \wedge g(b) \neq 0 \rightarrow$ divergira

• $\int_a^{\infty} \frac{g(x)}{x^s} dx$

$s > 1 \rightarrow$ konvergira

$s \leq 1 \wedge |g(x)| > m > 0$ za vse x od neke naprej
 $\rightarrow$ divergira

5. Ali konvergira?

a) $\int_0^1 \frac{1}{\sqrt{1-x^4}} dx =$

$1-x^4 = (1+x^2)(1-x^2) = (x^2+1)(x+1)(1-x)$

$= \int_0^1 \frac{1}{\sqrt{x^2+1} \sqrt{x+1} \sqrt{1-x}} dx = \int_0^1 \frac{((x^2+1)(x+1))^{-\frac{1}{2}}}{\sqrt{1-x}} dx$ $\leftarrow g(x)$ zvezna, omejena
 $s = \frac{1}{2} < 1 \Rightarrow$ konvergira

b) $\int_1^{\infty} \frac{\sqrt{1+x^2}}{x^3+1} dx = \int_1^{\infty} \frac{x^2 \sqrt{1+x^2}}{x^3+1} dx = \int_1^{\infty} \frac{g(x)}{x^2} dx$

$\lim_{x \rightarrow \infty} g(x) = 1 \Rightarrow \exists M: \forall x \geq M \quad g(x) \leq 2$

$s = 2 > 1 \Rightarrow$ konvergira

c) $\int_0^{\infty} x^{s-1} e^{-x} dx, s \in \mathbb{R}$

$\lim_{x \rightarrow \infty} \frac{x^{s+1}}{e^x} = \lim_{x \rightarrow \infty} \frac{(s+1)s \dots (s-k)}{e^x} = 0$

$= \underbrace{\int_0^1 x^{s-1} e^{-x} dx}_{\text{če } 1-s < 1} + \underbrace{\int_1^{\infty} x^{s-1} e^{-x} dx}_{\text{konv.}}$

$s > 0 \Rightarrow$ konv.

